

Quantitative Empirical Methods Written Questions

9:00AM August 22, 2025

Beniamino Green

Short Answer Question 1

Provide brief definitions of the following terms:

- a) p-value
- b) Sharp null hypothesis
- c) Stable Unit Treatment Value Assumption

- a) A p-value gives the probability of observing an event as or more extreme than the observed one under the null hypothesis. As an example, a p-value of .025 means that under the null hypothesis there is just a 2.5% chance of observing an event as extreme or more extreme than the one we observe in the data.
- b) A sharp null is generally used in causal inference, and refers to the hypothesis that a treatment under study has no effect for any unit ($Y_{i(1)} - Y_{i(0)} = 0 \quad \forall i$). This is the type of null hypothesis that is tested by a conventional randomization test.
- c) The stable unit treatment value assumption entails that a unit's outcomes under treatment and control are a function of **only** its treatment status, and not the treatment status of other units. This enables us to write the switching equation that relates potential outcomes to observed outcomes, $Y_i = Y_{i(1)}D_i + (1 - D_i)Y_{i(0)}$

Short Answer Question 2

Consider an experiment with $2n$ units, with n units receiving the treatment and n units receiving the control, and test the sharp null hypothesis using randomization inference at $\alpha = 0.001$.

- Explain why, in order to have $\alpha = 0.001$, we must have a design that yields at least 1,000 total randomizations (i.e., different ways to assign treatment).
- What is the minimum value of n for a matched pair experiment, so that the smallest p-value does not exceed than 0.001?
- What is the corresponding minimum value of n for a completely randomized experiment?

Part A:

This is because when we perform randomization inference, we generate p-values by re-randomizing treatment labels according to the design of the experiment, and seeing how many assignments yield statistics that are as extreme or more extreme than the observed statistic. We can then write the randomization inference p-value as:

$$p = \frac{\text{number of assignments giving as or more extreme outcome}}{\text{number of possible assignments}}$$

There is always one treatment assignment that generates a statistic as extreme as extreme as the observed one, namely the assignment that actually occurred. This means that the numerator must always be greater than or equal to one, and we can bound the smallest-possible p-value from below by:

$$p > \frac{1}{\text{number of possible assignments}}$$

In order to have an α value of .001, we must be able to generate p-values smaller than .001, so we can see that we will need more than 1000 possible assignments to get a value of α this small.

Part B:

Based on my answer to A, I need to find the smallest matched pair design that has over 1000 possible treatment assignments. The number of possible treatment assignments from a matched pair design is 2^n where n is the number of matched pairs. We can find this with $n = \lceil \log_2 1000 \rceil = 10$ so we have 10 pairs or 20 units.

Part C:

Based on my answer to part A, I need a design that has over 1000 possible treatment assignments. Because this design involves taking $2n$ units and selecting n of them for treatment, I am looking for the smallest n such that $\binom{2n}{n} > 1000$. By checking numbers in increasing order, I find that this happens when n is equal to 7 (7 units in the treatment and control arms for a total of 14 units).

Short Answer Question 3

Is statistical significance “transitive” in the sense that if A is statistically significantly different from B and B is statistically significantly different from C , A is statistically significantly different from C ? If Yes, explain why. If No, give a counterexample.

No, statistical significance is not transitive in this sense. Given the question setup, there is nothing preventing me from making A and C the same estimate with the same standard error, and setting B equal to a different estimator that is statistically distinguishable from A and C . A and C not significantly different from each other (as they are the same estimate), so this is a valid counterexample.

Short Answer Question 4

Diana Mutz writes in *Population-Based Survey Experiments*: “It is common practice in many fields of social science to reassure readers that they can trust the internal validity of a given experiment’s results by running so-called ‘randomization checks,’ that is, comparisons between various experimental groups on variables that are not part of the central theoretical framework of the study. Often, these are demographics . . . In general, a referee who suggests that an experiment requires a randomization check is probably well meaning, though wrong.” On the other hand, Alan Gerber and Donald Green write in *Field Experiments: Design, Analysis, and Interpretation* that “One way to bolster confidence in the integrity of the randomization procedure and the administrative care with which the data were handled is to provide a statistical description of the balance between treatment and control groups on a range of available covariates.” What are 2-3 arguments for conducting a randomization check? 2-3 arguments for not conducting a randomization check? Who is right? Why?

I think that I side more with Gerber and Green, but I understand Mutz’s perspective.

Mutz likely is emphasizing the fact that randomized experiments are balance observed and unobserved covariates *in expectation*, but not in any specific experiment. Balance tests are designed to test whether the population a treatment and control units come from different populations. If you are completely certain of the integrity of your experiment, you know that the two groups were drawn from the same population, so running a test to investigate this does not make sense.

Mutz might also point out that, imbalances that are not statistically significant still contribute to error. This is especially true for imbalances in variables that strongly determine the outcome. A small imbalance in such variables can greatly skew the results in an experiment, which means that balance tests are not a reliable way to rule out problematic imbalances between treatment and control arms. Observable imbalances can still be adjusted for, even if the imbalances are not statistically significant.

Green and Gerber are likely coming from a slightly different perspective. While randomized experiments do produce balance in theory. RCT’s are also difficult to implement, and there are many opportunities for randomization to fail. One of the practice exams includes an example about a mammography study where it was alleged that nurses were overriding the treatment assignment to try and shift high-risk patients into the treatment arm. Balance tests can help identify these kinds of subtle randomization failures that might otherwise be invisible to the trial administrators.

Balance tests can also help identify other factors that cause the treatment groups to become imbalanced after treatment was assigned and despite randomization. As an example, the investigator of a medical trial may be sure that the treatment was randomly assigned to patients. However, patients have a strong preference for the investigational drug over the existing standard of care, and often drop out of the trial if assigned to the control arm. In this case, even though the investigator correctly believes in the integrity of the randomization procedure, a balance test would help pick up on the fact that the treatment and control arms are no longer comparable because of attrition.

My perspective is a synthesis of the two positions. I agree with Mutz that if I were absolutely confident in the randomization procedure, it does not make sense to run a balance test after the fact. However, as Gerber and Green point out, there are many ways that randomization can fail in practice, and balance tests are also useful to identify factors like differential attrition that can bias study results. These considerations mean that I am generally in favor of performing balance tests for randomized experiments.

Short Answer Question 5

A rideshare platform enters cities ($n = 100$) sequentially from 2010-2014. Entry timing depends on: (1) city population thresholds, (2) regulatory approval processes of varying length, and (3) local taxi industry lobbying intensity. You want to estimate effects on employment in the local transportation sector. You have data on: (1) monthly city-level overall and transportation-specific employment from 2008-2016; (2) platform entry dates and usage volumes by city; (3) city characteristics including a measure on the strictness of taxi regulations, population, income, campaign finance, and election results; and (4) monthly city-level public transit ridership data from 2008- 2016. Note that some cities have very strict taxi regulations (e.g., caps on number of taxis allowed to operate in the city) while other cities have very lax regulations (e.g., no caps).

Design your most convincing identification strategy to estimate effects on rideshare platform entry on employment in the local transportation sector. What variation are you exploiting? State your key identifying assumption(s) and explain why they might be violated. How would you test for them given the available data?

A regression discontinuity based on city population size could be promising, but will not perform well with 100 observations and neglects the panel structure of the data. Additionally, there are generally multiple compound treatments associated with crossing a population threshold, which would make it very difficult to cleanly estimate the effect of the ride-share app's entry.

Because of this, I will try and use a panel approach that exploits the staggered rollout of the rideshare app to different cities. It sounds like treatment is not reversible, so I will use the multiperiod DID estimator developed by Callaway and Sant'Anna (2021). While this estimator can incorporate time-varying covariates, I would not try and control for the changes in public transport ridership, as these covariates are downstream of treatment and trying to adjust for them would mean 'adjusting away' some of the treatment effect that operates through these avenues.

The key identifying assumptions of my approach are SUTVA, parallel trends, and limited or no treatment assumption, and an overlap condition (Callaway and Sant'Anna 2021, § 2.2).

- The parallel trends assumption can be motivated by creating event study plots which display the trends of treatment and control groups before treatment and after differencing out unit and time fixed effects. If the two groups evolve similarly before treatment is assigned, then it becomes more plausible that they would continue to evolve in parallel after treatment is assigned (although this can never be proven).
- The no-anticipation assumption is more difficult to rule out, but can be relaxed slightly if we can specify a maximum length beyond which units cannot anticipate treatment (Callaway and Sant'Anna 2021, § 2.3). This could be plausible if approval for the rideshare app to operate in an area never takes more than a certain number of years, and it is impossible for citizens in an area to anticipate whether they will be chosen as the next candidate for expansion by the rideshare app.
- Overlap is normally testable, but overlap violations will be difficult to diagnose in a sample of 100 units.
- SUTVA violations would also be difficult to identify statistically, and would likely have to be justified based on domain knowledge. If cities are far enough apart from each other, it is more plausible that the economic effects of rideshare rollout do not influence outcomes in another cities. However, a subtle SUTVA violation does occur if cities make decisions on *how* rideshares can operate based on the observed outcomes of other units.

Short Answer Question 6

Scholars have debated the practice of regression adjustment to improve the precision of treatment effect estimates in randomized experiments. It is expected that you will use the internet to research this topic, and your answer should cite relevant readings and references.

- Critically evaluate several arguments for and against the use of regression adjustment with OLS for randomized experiments. (recommended length: about 300 words)
- Under what circumstances is regression adjustment with OLS approximately unbiased?
- Under what circumstances does regression adjustment with OLS improve precision?

Part A:

My understanding of the controversy surrounding using regression adjustment to adjust for imbalances between the treatment and control groups in randomized experiments is mostly informed by Freedman (2008)'s criticism of the practice, and responding papers by Green and Aronow (2011), and Lin (2013), and especially Chang et al. (2024), who provide an overview of debate in introducing a bias-correction method for covariate adjustment. I agree with perspectives taken in the response papers that while the objections to the practice are valid, they are either not applicable in large samples, or are relatively easy to remedy, so they do not force us to abandon the technique. These arguments considered, I then present another reason to avoid regression adjustment: the availability of methods that balance covariates by design, and methods that can adjust for more complicated functions of covariates estimated via machine-learning methods.

Freedman's Critiques and Responses

First, I lay out the objections raised by Freedman (2008, p 2), who raises four main arguments against the use of the multiple regression estimator for the ATE. Specifically, Freedman shows that the multiple regression estimator for the ATE is (1) biased in small samples, (2) in some cases will give greater asymptotic variance than the difference in means estimator, (3) can produce invalid confidence intervals in some settings, and (4) can even produce different confidence intervals than the difference in means estimator when the two return the same point estimate (Freedman 2008, p 2).

The first point appears only to be relevant in small samples. Freedman showed that the OLS estimator has a finite-sample bias that decays at an $\mathcal{O}(\frac{1}{n})$ rate (Freedman 2008, p 10), which means that in very large samples, this bias is likely to be negligible. In fact, in a response paper (Green and Aronow 2011), show that for even small sample sizes, the bias is generally so small to be negligible in even quite small sample sizes ($n=15$). Additionally, in a recent paper (Chang et al. 2024) have also derived a formula for correcting for this bias, at the cost of a slight increase in variance.

The remaining points are real concerns, but only apply in some situations, and are easily corrected for (Lin 2013). Specifically (Lin 2013) shows that covariate adjustment never hurts the asymptotic variance of the estimator if all covariates are interacted with treatment (Lin 2013, p 301).

Additionally, the sandwich estimator consistently estimates the variance of the multiple regression estimator (Lin 2013, p 306), which addresses points 3 and 4. Both of these facts are emphasized by Chang et al. (2024).

Alternatives to Regression Adjustment

Ultimately, I believe the most compelling reason not to use regression adjustment in RCTs is the availability of other methods that can perform balancing at the design stage, or balance on more complicated functions of covariates estimated by machine learning.

Instead of adjusting for balances post-hoc, researchers who have access to covariates when treatment is assigned can pick an assignment that minimizes imbalances between treatment and control groups, such as a matched-pair experiment. A recent advance in this literature is the Gram-Schmidt-Walk (GSW) design, which achieves a ‘regression by design’ approach, meaning the variance of the Horvitz-Thompson estimator under the GSW design is bounded by the loss of a penalized regression of the sum of potential outcomes on the covariates (Harshaw et al. 2024, p 7).

Even when pretreatment covariates are not available, researchers might want to consider other forms of covariate adjustment that can account for more complex imbalances between the two groups. As an example Middleton and Aronow (2015) derive an estimator for the average treatment effect based on the propensity score that is unbiased in finite samples even when covariate adjustment is misspecified. This estimator allows us to use machine learning and cross-fitting to estimate the ideal covariate adjustment function, and then use it to adjust away imbalances in a data-adaptive manner.

Part B:

Because the bias of the OLS estimator decays with at a $O(\frac{1}{n})$ rate, the OLS estimator will be approximately unbiased whenever n is large. There are also some corner cases in which the OLS estimator is exactly unbiased, such as (trivially) when there are no treatment effects for any unit, or when treatment effects are constant and the number of treated and control units is identical (Green and Aronow 2011, p 6).

Part C:

Covariate adjustment will help precision in cases where covariates strongly predict outcomes, and the number of covariates is small relative to the sample size. In these cases, most of the variation in the outcome can be residualized out by the adjusting regression, allowing for a lower-variance estimate of the causal effect.

Short Answer Question 7

You are asked to review a paper that is examining the impact of algorithmically generated risk recommendations to judges' sentencing decisions on whether or not to impose cash bail for individuals in pre-sentencing trials. The algorithm is deterministic, and takes in a risk score that ranges from 1 to 6, where 1 is considered low-risk and 6 is high-risk. A risk score above a 3 will result in the algorithm recommending cash bail, whereas a risk score below a 3 will result in the algorithm recommending release. The authors want to know whether a "cash bail" recommendation from the algorithm will change the sentencing decision of the judge. The outcome of interest here is the sentencing decision of the judge, where $Y = 1$ for sentencing with cash bail and 0 implies release.

- a) The authors begin by using a sharp regression discontinuity design. They report a treatment effect of 0.3. What is the estimand of interest in an RDD? What target population does the RDD estimate for?
- b) Explain in words how you would run this RDD. What is the running variable?
- c) What assumptions must be true for this to be a valid RDD?
- d) Explain the different researcher degrees-of-freedom that can impact the downstream analysis.
- e) The authors do not provide any robustness checks. What are some robustness checks the authors should provide to help reason about the validity of their underlying RDD assumptions? Name 2 examples of robustness checks, and how they would help improve the credibility of their RDD estimate.
- f) In the next part of the paper, the authors run a regression, where they regress the judges' sentencing decisions with the algorithmically generated risk recommendations, and a set of pre-treatment covariates, such as the demographic information about the cases in the study and judges fixed effects. They find a statistically insignificant treatment effect, with point estimate of 0.2. They conclude that even though it is not significant, the point estimate is positive, and validates the RDD result. Is this a valid claim? Explain why or why not.

Part A:

The sharp RDD estimates the effect of recommending cash bail on the judge's decision to release or impose bail on a marginal defendant who has a risk score of 3. This is a local average treatment effect, and applies to units at the cutoff value of 3.

Part B:

There are a few different methods to estimate RDD's, all of which operate by estimating the limit of the control outcome approaching the cutoff point from the left and subtracting this from an estimate of the right-hand limit of the treatment outcome at the control point. If I were to do this, I would use the `RdHonest` package in R, which performs a local linear regression on each side of the regression, using a weighting kernel with a bandwidth that is chosen to minimize the width of the confidence interval associated with the estimate (Kolesár 2020).

Part C:

The following assumptions are needed to run an RDD:

- Some units within a neighborhood of the cutpoint with probability 1
- The conditional expectation functions $E[Y(0)|x]$ and $E[Y(1)|x]$ are continuous at the cutoff point.

Of note is that the second assumption rules out compound treatments, which are very common in practice. If there are several different treatments that are applied at a risk score of three (such as increased security in the courtroom), then these effects cannot be separated.

Part D:

Generally, there are several degrees of freedom that the researchers have in implementing the RDD design. They could choose between several estimation approaches based on local regressions, which differ in how they choose the smoothing parameter (`RdHonest` and `rdrobust`). Even holding the specific implementation constant, most software packages for RDD allow you to choose the type of kernel you wish to use to fit the regression (triangular, rbf, square, etc).

Researchers also have the classic problem of choosing which covariates to adjust for, which quickly expands the number of possible specifications. Finally, researchers who fit a global polynomial regression to data on each side of the cutoff have a great deal of flexibility in choosing the order of the polynomial on each side of the cutoff, as highlighted by Gelman and Imbens (2018).

Part E:

There are several robustness checks that the authors can perform which will help motivate the RDD assumptions.

One RDD assumption is that the potential outcome functions are continuous at the cutpoint. This can't be directly tested, but one can help motivate this by showing that the observed outcomes are continuous at regions away from the cutpoint using a series of placebo cutpoint tests. This will not be a reliable indicator if there are other treatments that happen at other values of the running variables, but continuity away from the cutpoint can help motivate the assumption that there is continuity at the cutpoint.

Another way that the potential outcomes can be discontinuous at the cutpoint is if there is sorting around the cutpoint. The McCreary density test can be used to look for a discontinuity in the density of observations around the cutpoint. If there is a discontinuity, this suggests that sorting may be taking place and the continuity assumption is likely not met.

Placebo outcomes can also be used to test for sorting if these are collected before treatment is assigned. If treatment assignment can be shown to influence pre-treatment characteristics such as demographics, this is another sign that sorting may be taking place and thus that the conditional expectation functions are discontinuous at the cutpoint.

Part F:

I do not find this claim convincing. First, the estimate is not significant and a positive sign will happen roughly 50% of the time under the null hypothesis, so this is not strong evidence of an effect.

These points aside, the regression that the authors employ will give a consistent estimate of the effect of the algorithmic recommendation under conditional ignorability, which demands that potential outcomes are independent of treatment, conditional on the observed covariates. It is likely that there are unobserved factors about the case that influence both the judge's decision and the risk recommendation which would confound the association between treatment and the outcome. Because of this, I do not believe that this regression estimates a causal quantity.

Essay Question

Read the following article and offer a critical evaluation of its methodological approach and presentation of results. Note: “critical” does not imply that you must only criticize – it is recommended that you give credit to the authors if and when their arguments are convincing and/or novel with respect to standard practice. Your response should be between 500 to 1000 words. We advise you to pay particular attention to the identification conditions (either explicit or implicit), the associated estimation strategy, possible threats to inference, statistical power, and generalizability. Justify each of your claims and, where applicable, suggest ways in which this line of research might be improved. We do not expect you to have special expertise in the topic area, but we do expect you to bring to bear your general analytical skills as a political scientist.

Article:

Myers, Andrew C.W. “Press Coverage and Accountability in State Legislatures.” *American Political Science Review* (Forthcoming). <https://doi.org/10.1017/S000305542500022X>

Press Coverage and Accountability in State Legislatures (Myers 2025) sheds light on the effect of media coverage of local politicians on several political important outcomes including legislator name recognition, incumbency effects, and ballot roll-off. Because the relationship between coverage intensity and the outcomes under study is confounded by ambient cultural and economic factors, the paper uses an element of newspaper intensity that is plausibly exogenous from these factors: variation in coverage of legislators induced by the congruence or discongruence between newspaper markets and congressional districts.

I think this paper investigates an important question, and I appreciate the detailed examination of multiple outcomes downstream of media coverage. In my response, I discuss my reaction to two of the analysis decisions in the the paper. Specifically, I worry that the measure of the independent variable is not validated and may fail to capture the true patterns of congruence on the ground. Second, while the paper is based on the assumption that congruence is exogenous, no formal statistical tests are deployed to help motivate this assumption, and I am skeptical that the expansion decisions of newspaper operators, and the subscription patterns of their readers are exogenous of salient political and cultural factors that influence the outcomes under study.

Measurement of Congruence

First, I discuss the measurement strategy used to quantify the extent of newspaper-district congruence in the paper. My worry is that the measure may not accurately measure congruence as the data on newspaper circulation is collected at the county level, and then used to construct district-level treatment variables in a process that is not validated against an external measure of congruence or newspaper readership.

The paper crosswalks county data to legislative-district circulation under a proportionality assumption which implies that if a congressional district divides a county in half, then any newspaper’s readers will also be evenly divided between the two legislative districts. This assumption is crucial in the context of a paper that leverages variation in local circulation patterns for identification, but it is never interrogated. Given that many newspapers will service specific towns or regions of a county, which might also line up with legislative district, I think this could be false in many cases. If this assumption is not reliable, then the measurement of the key independent variable may be biased, and the treatment effect estimates will not be reliable.

I am also worried that a large percentage of the data used as inputs to this calculation could have been imputed rather than directly measured. The paper describes the data collection process as follows:

“I calculate Congruence for every district in all 98 partisan state legislative chambers for the years 2000–2022 ... using county-level newspaper circulation data from Peterson (2021a)¹⁰. these data were digitized from the 2008, 2014, and 2018 editions of the Standard Rate and Data Service Circulation handbook¹¹.” (Myers 2025, p 4)

The last footnote in the paragraph notes that the data for missing years is imputed. I think that these statements might unintentionally misdescribe the dataset because the referenced paper, *Not Dead Yet* by Peterson (2019), describes the dataset as covering the “United States in 2008 and 2014,” and as constructed using “editions of the Standard Rate and Data Service Circulation handbook in these years” (Peterson 2019, p 345). The fact that the citations disagree on the specific editions of the handbook used would ordinarily be trifling, but it is notable because according to the Library of Congress, page makes clear that this handbook stopped being released in print 2014 when the publisher moved to a subscription business model (Library of Congress, n.d.).

This makes me uncertain about how much of the data is imputed based on prior years and trends. If small portions of the data are filled in by imputing relatively static circulation numbers, this is clearly not a concern. However, I would find the results of the paper less credible if the main explanatory variable is based on measurements taken at two points in time (2004 and 2008), which are then extrapolated forwards and backwards in time to cover the rest of the dataset.

I suggest that the paper could be improved by providing more information on the extent to which the data was imputed, and a series of validation tests showing that the congruence measurement is actually reflective of newspaper diets on the ground. Specifically, I would like to understand the fraction of the data that was imputed in the main analysis and the method used for imputation, especially if data is filled-in based on extrapolating sales in newspaper trends outside of the support of the observed data. Additionally, it would be valuable to see the extent to which this measure of congruence aligns with any other established measures (if any exist), or if there is a way to verify that the newspaper market share figures calculated for each congressional district are reflective of local news consumption.

Conditional Ignorability

I now turn to discussing the conditioning-on-observables strategy used by the paper to estimate the effects of media congruence, and the conditional ignorability assumption required to generate valid causal estimates in this setting.

The regression design used in the paper returns valid causal estimates if, after linearly adjusting for imbalances between the control and treatment groups, potential outcomes are independent of treatment assignment. This assumption is defended in the paper using a qualitative examination of a map of congruence levels across the state of Wisconsin. The fact that neighboring districts in this map can often have drastically different levels of congruence is taken as evidence that congruence is exogenous (Myers 2025, p 4), and so the high-congruence and low-congruence populations are plausible counterfactuals for each other after regression adjustment.

My worry is that this examination is performed in lieu of a formal statistical test for balance, and detailed qualitative examination of coverage variation may not be able to detect subtle confounding patterns which operate at the level of averages. Just because there is substantial variation in treatment status does not mean that treatment variation is completely exogenous: it could be that

there is a small portion of treatment variation that is systematically explainable by confounding factors which are difficult to detect with the human eye but will still meaningfully bias results.

Indeed, I believe that there likely are imbalances between the treatment and control units, and that these imbalances cannot be adequately adjusted for using linear regression: the decision of a newspaper to expand or contract its distribution is based on complex weighing of multiple competing considerations including the cost of expanding, demand for the paper in the new region, and competition from competitors, and these same factors are likely correlated with the outcomes under study. To give an example as an example, the large population of big cities means that they can likely to sustain multiple newspapers. Big cities are also more likely to be near population congressional district boundaries, especially if city is so large as to require several congressional districts. These phenomena would naturally lead to a situation in which highly-urban congressional districts are overrepresented among low-congruence districts. To control for this mechanism, the author controls for the size and urban composition of the congressional districts in their analysis. However the worry is that there are other similar imbalances that cannot not be adjusted for because the relevant information was never recorded in the dataset to begin with.

In this context, a balance test that compares high and low-congruence districts could help persuade readers that the treatment and control groups are indeed comparable and are less likely to differ in terms of unobservables that cannot be adjusted for. Additionally, if there are imbalances between the two groups a set of sensitivity analyses could help persuade readers that any unobserved confounders would have to explain an unreasonable amount of the remaining variation in the outcome in order to reverse the association seen in the data. These approaches would allow readers to have more confidence in the conditional ignorability assumption, or show that violations of this assumption are likely to have small effects, bolstering confidence in the results of the paper.

Discussion

Press Coverage and Accountability in State Legislatures does much to advance the literature on the effects of media coverage on local political outcomes using a creative identification strategy that uses plausibly-exogenous variation in media coverage induced by newspaper-district congruence. As the paper notes, understanding the effects of local news coverage is becoming increasingly important in a context of declining local newspaper readership. In my response, I have highlighted the difficulty of conducting causal inference in this setting, which requires both precise measurement of newspaper congruence at the congressional-district level, and successfully isolating intensity of press coverage from confounding factors. I believe the paper makes good arguments to address both of these concerns, but I suggest that the argument of the paper could be made stronger by more transparent discussion of the measurement process and imputation strategy, as well as more balance tests to convince readers that newspaper congruence is completely exogenous from other confounding factors.

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Quantitative Empirical Methods Coding Questions

9:00AM August 22, 2025

Beniamino Green

Computer-Aided Section

Generate Data Function:

Below, I include my `generate_data` function, which generates the dataset as described in the problem setup.

```
inv_logit <- function(x) {exp(x)/(1+exp(x))}

generate_data <- function(n) {
  # Generate the continuous and discrete covariates
  X_cont <- matrix(rnorm(n*5), nrow = n, ncol = 5)
  X_discrete <- matrix(rbinom(n*5,1,.5), nrow = n, ncol = 5)

  # Create X matrix by binding continuous and discrete together
  X <- cbind(X_cont, X_discrete)

  # Treatment probabilities and assignment
  pi <- inv_logit(X %*% c(1,1,0,0,0,0,0,0,1,0))
  z <- rbinom(n, 1, pi)

  # Generate Potential Outcomes
  beta <- c(1, 2, -.3, -.4, 1, 0.9, 2, -.1, 2, 1)
  y_0 = c(X %*% beta) + rnorm(n)
  y_1 = y_0 + z*2

  # Switching Equation
  y = ifelse(z, y_1, y_0)

  # Return Dataframe
  return(tibble(
    y = y,
    y_1 = y_1,
    y_0 = y_0,
    z,
    as.data.frame(X)
  ))
}
```

Code for Estimators:

In this section, I include my functions for estimating the treatment effects from the data. The `digm` function is responsible for taking the difference in group means on a dataset, while the `ols` and `ipw` estimate the regression and IPW estimators respectively. Both of these functions take a second argument, `variant` which determines the regression used in fitting the response or propensity surfaces.

```
# Calculates Difference in Group Means
digm <- function(df, variant = NULL) {
  z <- as.logical(df$z)
  return(mean(df$y[z]) - mean(df$y[!z]))
}

# Calculates OLS Estimate of Treatment Effect
ols <- function(data, variant = "a") {
  formula <- switch(variant,
    a = y ~ z + V1 + V2 + V3 + V4 + V5 + V6 + V7 + V8 + V9 + V10,
```



```

      b = y ~ z + V1 + V2,
      c = y ~ z + V1 + V2 + V9,
    )

    model <- lm(formula, data = data)

    return(coef(model)['z'])
}

# Calculates IPW Estimate of Treatment Effect
ipw <- function(data, variant = "a") {
  formula <- switch(variant,
    a = z ~ V1 + V2 + V3 + V4 + V5 + V6 + V7 + V8 + V9 + V10,
    b = z ~ V1 + V2,
    c = z ~ V1 + V2 + V9,
  )

  p_score_model <- glm(formula, family = binomial(), data = data)

  pi_hat <- predict(p_score_model, type = "response")

  z <- as.logical(data$z)

  treat_ipw_mean <- sum(data$y[z]/pi_hat[z]) / nrow(data)
  ctrl_ipw_mean <- sum(data$y[!z]/(1-pi_hat[!z])) / nrow(data)

  return(treat_ipw_mean - ctrl_ipw_mean)
}

```

Simulation Section

In this section, I use the previously defined function to create a tidy dataframe of the simulation results. The dataframe has one row for each method-simulation pair, or 7000 rows in total. Computing all the simulations up-front is a little memory-intensive, but it means that I can express the simulation logic in a clean, functional style. For a larger set of simulations, it would be possible to compute the simulations lazily, which would decrease memory usage.

```

# First, create a dataframe that has one row per estimator

estimators_df <- tribble(
  ~ method, ~ estimating_fn,
  "OLS", ols,
  "IPW", ipw,
)
estimators_df <- crossing(estimators_df, variant = c("a", "b", "c"))

estimators_df <- bind_rows(
  estimators_df,
  tribble(
    ~method, ~ estimating_fn,
    "DIGM", digm,
  )
)

```

```

# Then simulate 1000 datasets for each sample size
sim_df <- crossing(
  n = c(100, 1000, 5000, 10000),
  id = 1:1000
) %>%
  mutate(
    data = map(n, generate_data)
  )

# Join the estimators dataframe with the simulations dataframe,
# and compute each estimator on the dataset.

output_df <- crossing(estimators_df, sim_df) %>%
  mutate(
    estimate = pmap_dbl(list(estimating_fn, data, variant), \(x,y,z){x(y,z)})
  )

head(output_df)

```

```

## # A tibble: 6 x 7
##   method estimating_fn variant      n    id data                estimate
##   <chr>    <list>      <chr>  <dbl> <int> <list>                <dbl>
## 1 DIGM    <fn>          <NA>   100     1 <tibble [100 x 14]>    4.53
## 2 DIGM    <fn>          <NA>   100     2 <tibble [100 x 14]>    4.19
## 3 DIGM    <fn>          <NA>   100     3 <tibble [100 x 14]>    4.10
## 4 DIGM    <fn>          <NA>   100     4 <tibble [100 x 14]>    4.25
## 5 DIGM    <fn>          <NA>   100     5 <tibble [100 x 14]>    3.99
## 6 DIGM    <fn>          <NA>   100     6 <tibble [100 x 14]>    4.05

```

Results and Discussion

```

make_table <- function(simulation_df, n) {
  simulation_df %>%
    dplyr::filter(n=={n}) %>%
    group_by(method,variant) %>%
    summarize(mean = mean(estimate), sd = sd(estimate), MSE = mean((estimate-2)^2))
}

```

Summary Table for n=100

```
knitr::kable(make_table(output_df, 100), digits=2)
```

method	variant	mean	sd	MSE
DIGM	NA	4.46	0.56	6.39
IPW	a	2.02	1.85	3.42
IPW	b	2.53	1.23	1.79
IPW	c	2.02	1.82	3.30
OLS	a	2.01	0.26	0.07
OLS	b	2.47	0.52	0.49
OLS	c	2.00	0.48	0.23

Summary Table for n=1000

```
knitr::kable(make_table(output_df, 1000), digits=2)
```

method	variant	mean	sd	MSE
DIGM	NA	4.49	0.19	6.23
IPW	a	1.99	0.46	0.21
IPW	b	2.50	0.33	0.36
IPW	c	2.00	0.47	0.22
OLS	a	2.00	0.08	0.01
OLS	b	2.48	0.17	0.26
OLS	c	2.00	0.15	0.02

Summary Table for n=5000

```
knitr::kable(make_table(output_df, 5000), digits=2)
```

method	variant	mean	sd	MSE
DIGM	NA	4.50	0.08	6.23
IPW	a	1.99	0.18	0.03
IPW	b	2.50	0.14	0.27
IPW	c	2.00	0.19	0.04
OLS	a	2.00	0.03	0.00
OLS	b	2.48	0.07	0.24
OLS	c	2.00	0.07	0.00

Summary Table for n=10000

```
knitr::kable(make_table(output_df, 10000), digits=2)
```

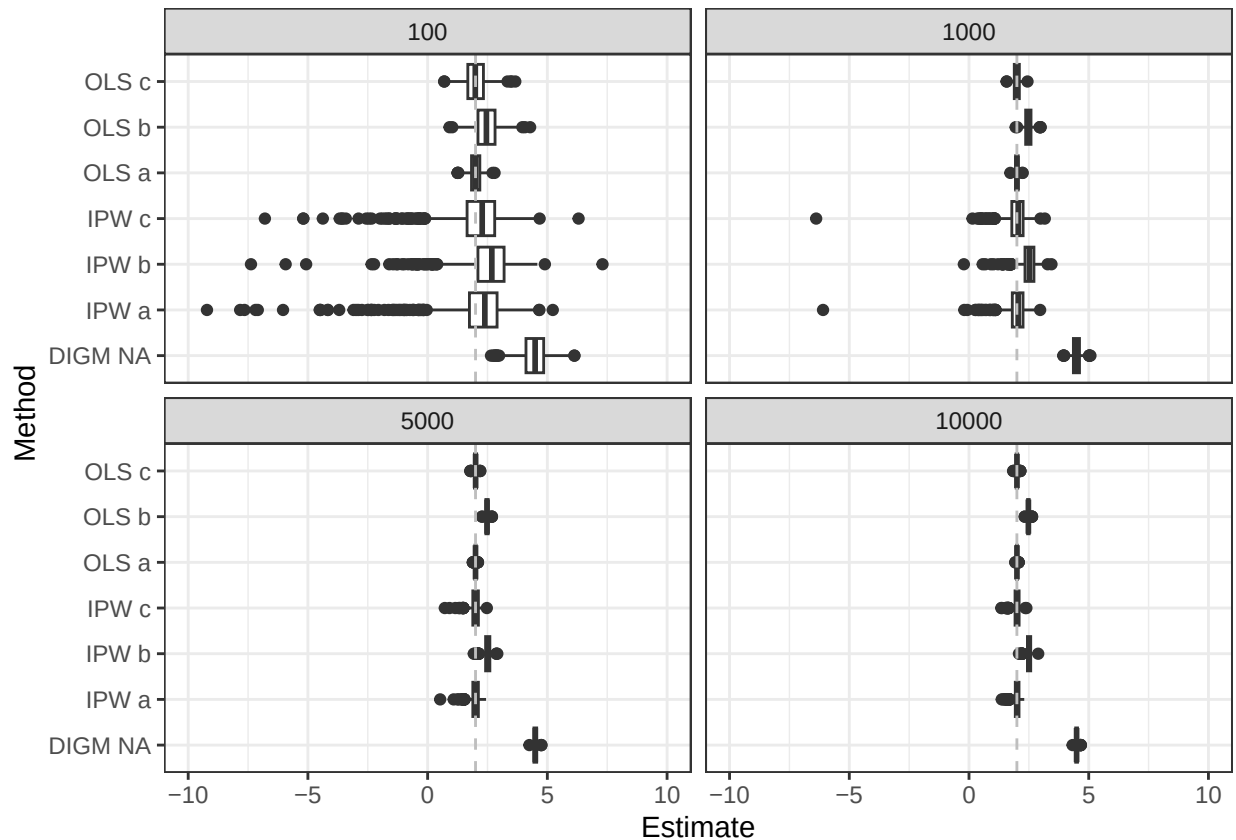
method	variant	mean	sd	MSE
DIGM	NA	4.49	0.06	6.22
IPW	a	2.00	0.13	0.02
IPW	b	2.50	0.10	0.26
IPW	c	2.00	0.13	0.02
OLS	a	2.00	0.02	0.00
OLS	b	2.48	0.05	0.23
OLS	c	2.00	0.05	0.00

Boxplots

Below I show a series of boxplots summarizing the distributions of estimates produced by each estimator:

```
ggplot(output_df, aes(x=paste(method, variant), y=estimate)) +  
  geom_boxplot() +  
  facet_wrap(~n) +  
  xlab("Method") +  
  ylab("Estimate") +  
  coord_flip() +
```

```
ylim(c(-10,10)) +
geom_hline(yintercept = 2, linetype = 2, col = "grey") +
theme_bw()
```



Discussion

In this exercise, I have estimated the performance of three different estimators. I summarize the results of this exercise below:

First, we can see that difference in group-means estimator is not consistent for the average treatment effect because there are confounding factors that influence both treatment and outcome. It seems to converge to 4.5, which reflects *both* the average treatment effect and a selection bias because units with higher outcomes are more likely to get treated.

The IPW estimate is always less stable than the OLS estimate based on the same variables. The IPW estimator has a higher-variance because propensity scores near zero and one mean that the estimate is very sensitive to the outcomes of a small fraction of the sample. Additionally, OLS calculates a precision-weighted average of the treatment effect which is an easier target than the ATE. In this simulation this is the same as estimating the ATE, but if we had treatment heterogeneity and heteroskedasticity, the OLS estimate would return a lower-variance estimate which is biased for the ATE.

We can see that specifications “A” and “C” yield consistent estimates of the average treatment effect as they are able to adjust for all the confounding variables. The model in specification B is not correctly specified as it does not include covariate 9 which influences both treatment and outcome. Accordingly, model B does not produce a consistent estimate of the ATE.

Finally, we can see that when using OLS, specification A has the lowest error. This is because including the

extra covariates allows us to adjust away chance imbalances between the treatment and control arms. The picture is less clear for the IPW estimators, and I am ambivalent as to which specification is better. The propensity scores estimated on all covariates will balance the means of these covariates in the sample, but they will also make overlap worse, which increases variance. It appears that neither principle is dominating here.